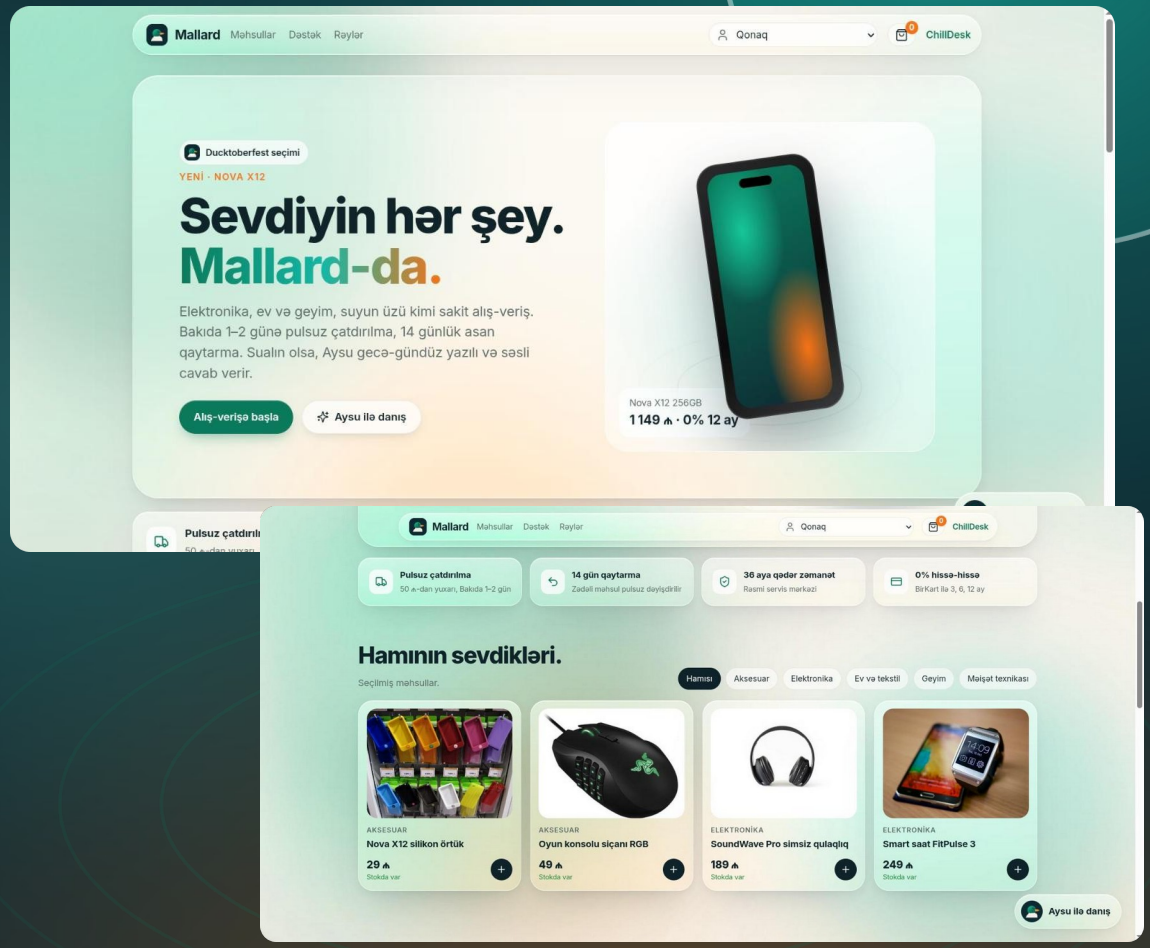
ChillDesk

An open-weight Gemma 4 AI support agent for online stores

Best Use of Gemma 4 | **Best Open-Source AI Project**



team-Chill · Ducktoberfest Baku 2026 · Hacktoberfest Hack Day

Support is slow, costly and scattered



24/7 questions

“Where is my order?”, “when do I get my money back?” The same questions repeat day and night while customers wait in a queue.



Manual checks

A damaged-product photo, a phone call, a receipt: an operator checks each one and decides by hand.



Manual wrap-up

Summary, category, ticket and a follow-up email take minutes after every conversation.

A small store can’t afford a 24/7 support team.

Meet Aysu

Chat that takes action

Gemma 4 function calling: finds and tracks orders, opens returns and tickets

Sees photos

Damaged-product photo → Gemma 4 vision → decision

Voice calls, 3D Aysu

Lip sync and tone-driven emotion, in Azerbaijani

Automation

Summary, priority, auto-ticket, follow-up email



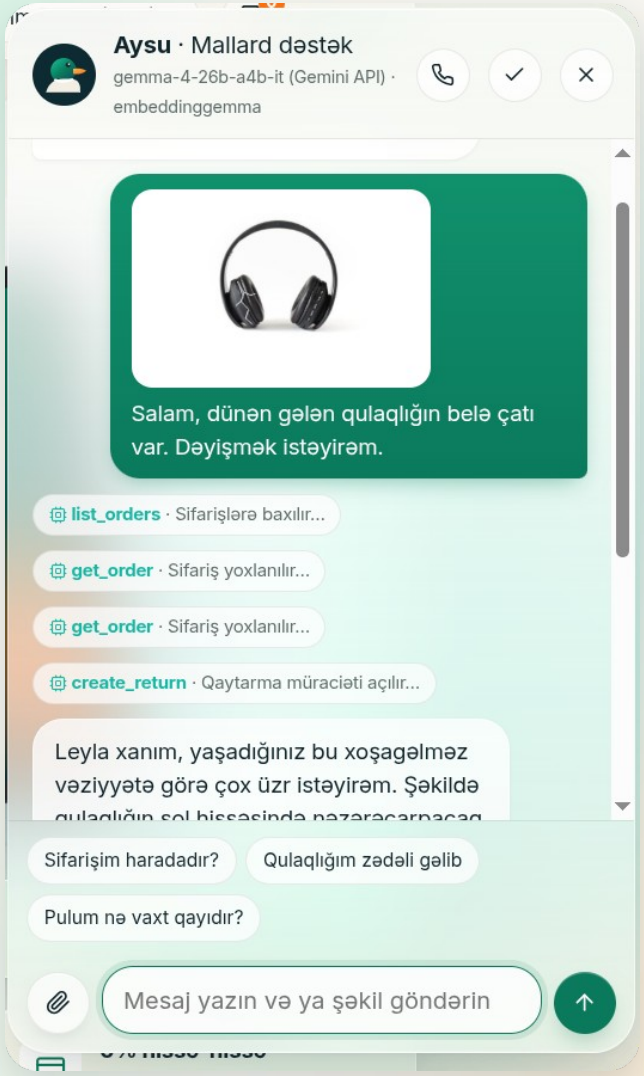
Aysu · Mallard support agent

Send a photo, get it solved



The photo the customer sent

≈13 seconds · 4 tool calls · 0 operators



1. **Gemma 4 reads the photo**

   “crack on the left ear cup” → photo_assessment

2. **list_orders → get_order**

   Finds Leyla’s order ML-1042 in Neon

3. **create_return**

   Policy engine: 14 days + damage = free replacement

4. **Answer + return RT-...**

   A courier swaps the old one for a new one

Just like a call: voice Aysu



The brain is Gemma 4

ElevenLabs does only STT/TTS; our Custom LLM endpoint writes the answer

Character-level lip sync

Per-character timing of every audio chunk + real loudness

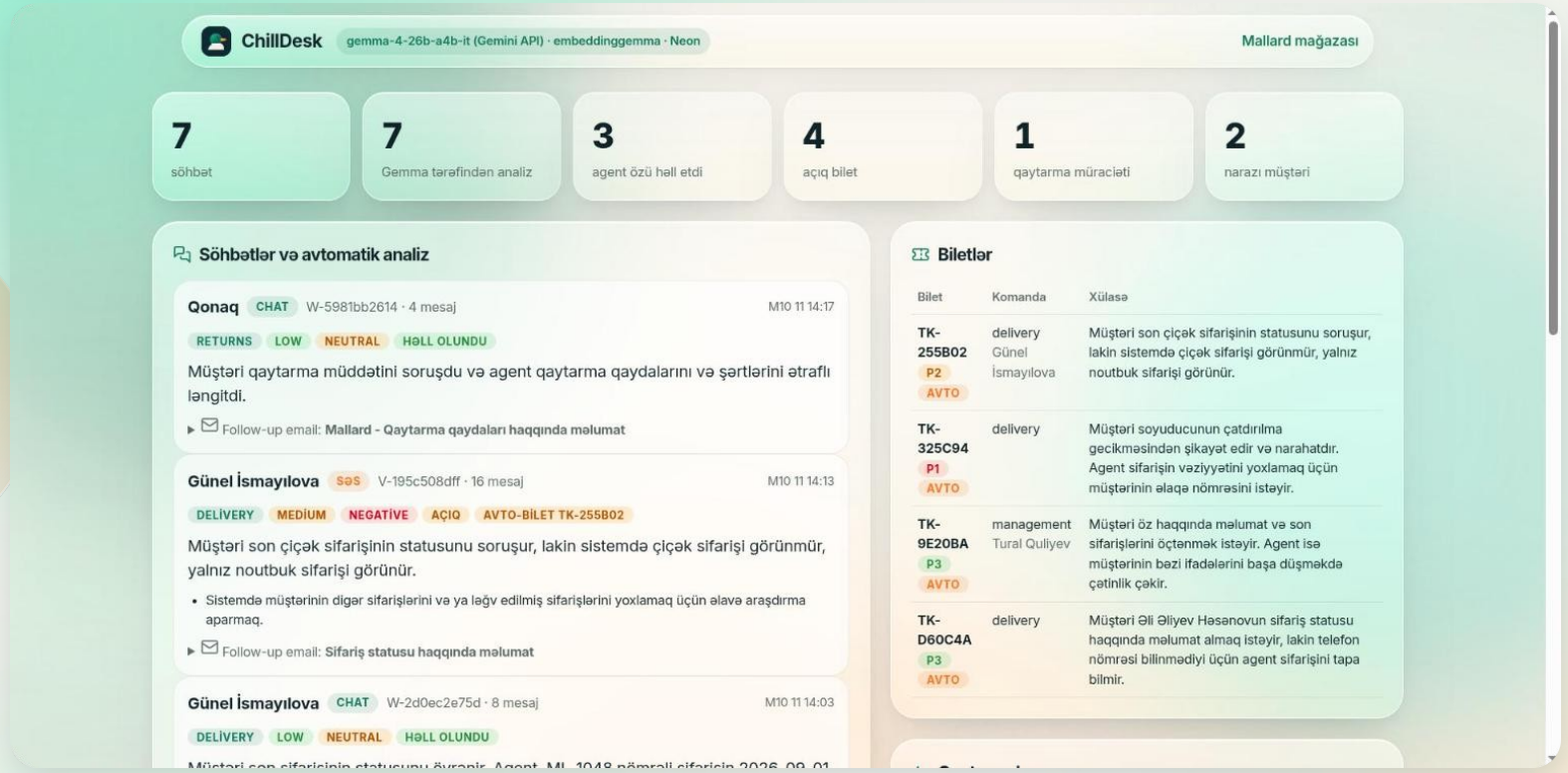
Emotion follows the tone

[empathetically] changes both the voice and the face

Feels like a real call

425 Hz ringback, Aysu appears when she picks up

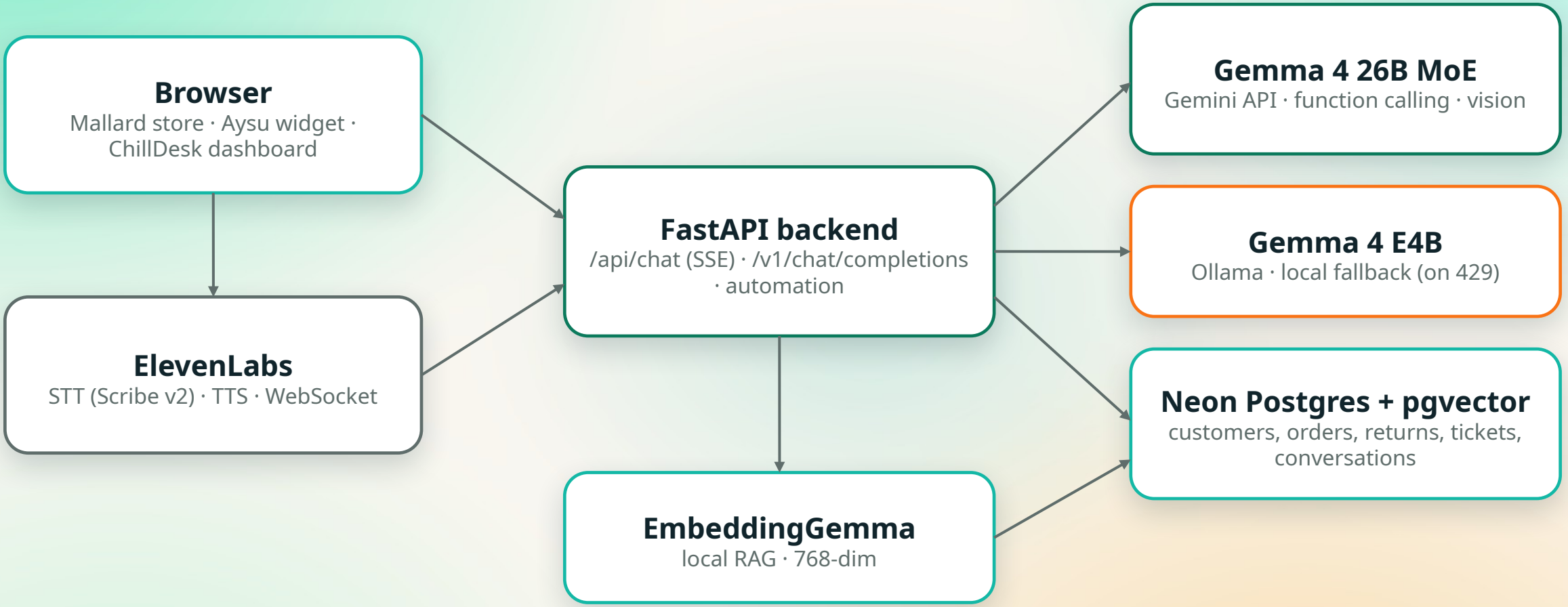
The chat ends, the work gets done



Gemma 4 → JSON

- Summary
- Category
- Urgency
- Sentiment
- Resolved?
- Needs a human?
- Auto-ticket
- Follow-up email
- KB gap

Architecture



We trained our own model: Unsloth QLoRA

229
labelled Azerbaijani conversations

1B
parameters — Gemma 3 1B (4-bit)

8 GB
laptop GPU (RTX 4070)

284 s
training time

1 · Data

We sample the labels first (7 categories × 3 urgencies × 3 sentiments, 12 products, 6 customer styles), then Gemma 4 26B writes a matching conversation. The ground truth is exact.

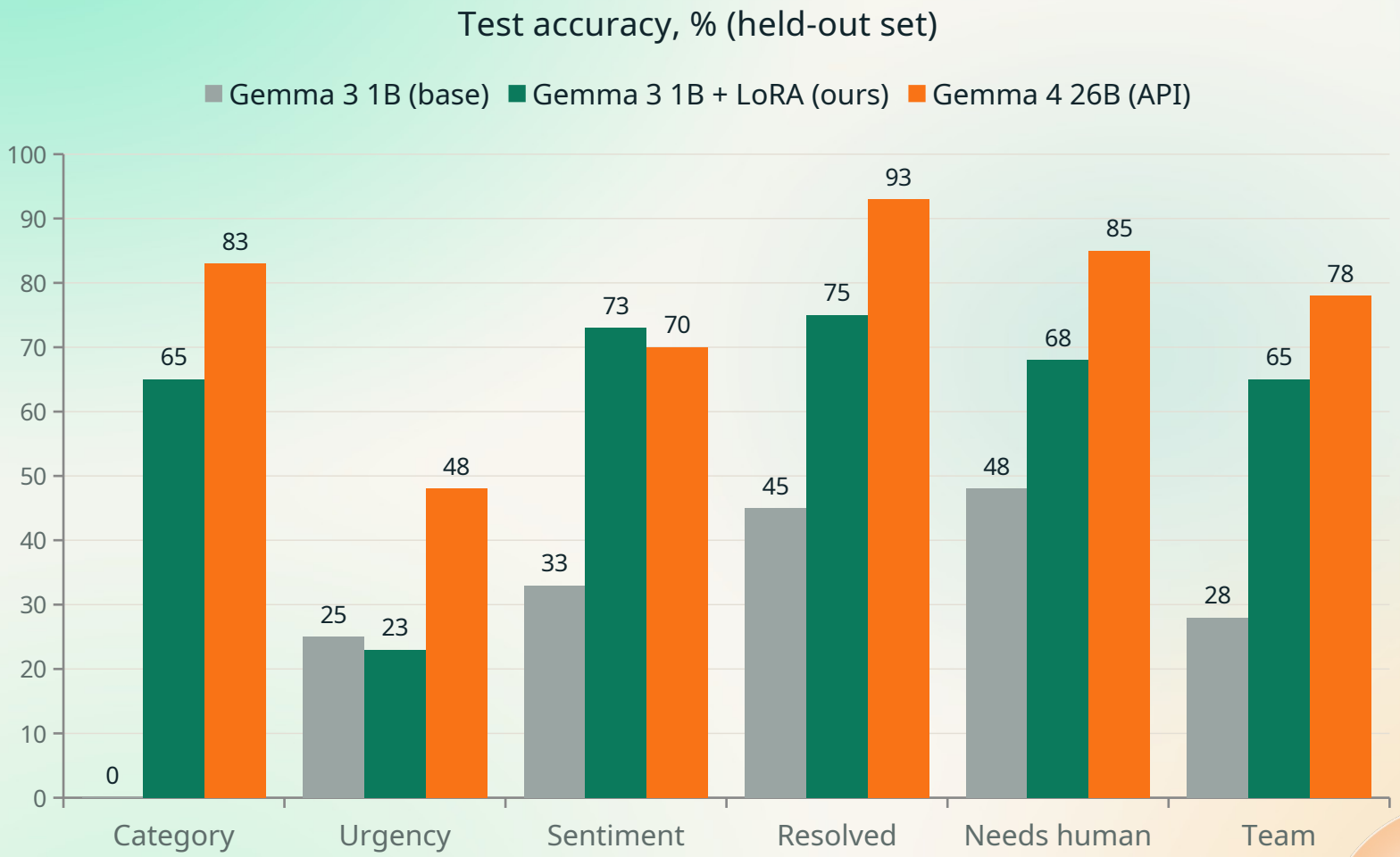
2 · Train

Unsloth QLoRA, rank 16, loss on the JSON answer only. The small model learns our labelling rules.

3 · Eval

40 held-out conversations: base 1B vs fine-tuned 1B vs Gemma 4 26B. Every field is scored separately.

Result: small model, big jump



61%

fine-tuned 1B mean accuracy

30%

same model, not fine-tuned

76%

Gemma 4 26B, zero-shot

Open all the way down

Gemma 4 26B / E4B

Agent, vision, analysis

Apache 2.0

EmbeddingGemma

Local RAG embeddings

Gemma Terms

Gemma 3 1B + LoRA

Fine-tuned analyst

Gemma Terms

Unsloth · Ollama

Training and local inference

Apache 2.0 · MIT

Neon · pgvector

Postgres + vector search

Apache 2.0 · PostgreSQL

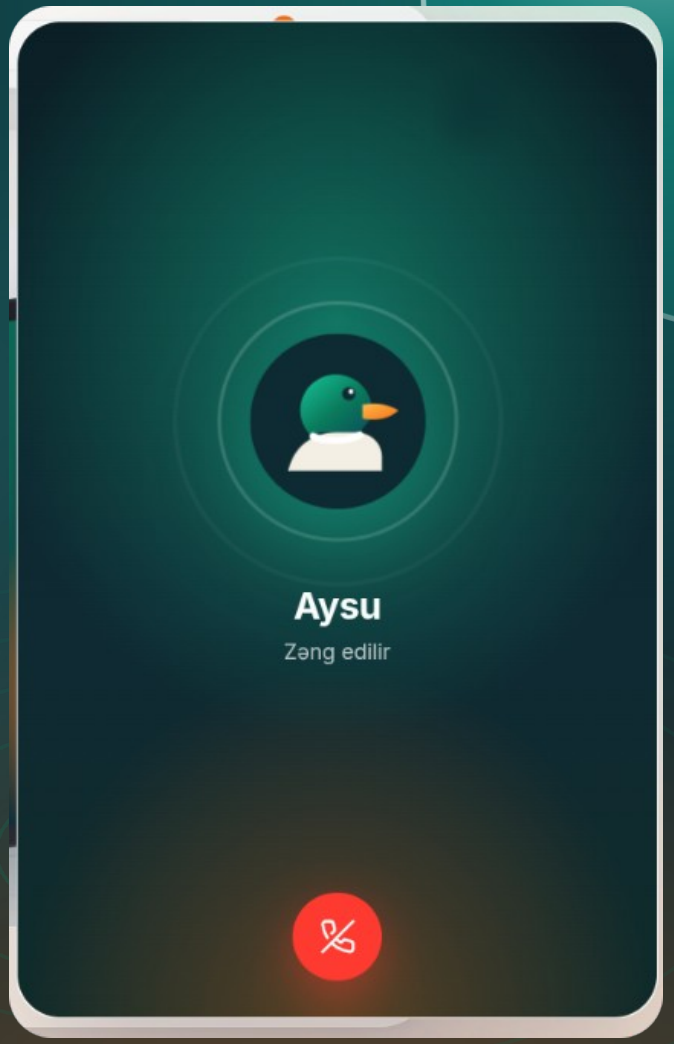
ChillDesk code

github.com/Ducktoberfest/team-Chill

MIT

Thank you!

Now let’s call Aysu.



github.com/Ducktoberfest/team-Chill